

Prediction Is Not Fact

Project: Helpful Classroom Sorter

1. Prediction Is Not Fact

Teach the learning cycle

Primary target: Separate a recorded prediction from the observation used to check it

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Boundaries and 'Not Sure'

Without opening the old lesson, explain one key idea from Boundaries and 'Not Sure'. Give an example and a mistake that the idea helps you avoid.

Look for: Identify a specific source of uncertainty and explain how it affects a prediction

Reference example, not the only acceptable wording: The rule and two stated possibilities are known; the actual height is unresolved. At 5 predict short, at 6 predict tall. Inspect the original readable record or remeasure. Repeating a calculation supplies no new information about which height was recorded.

Build a Human Classifier

Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

Look for: Construct a classifier with a stated input format, selected rule and recorded output

Reference example, not the only acceptable wording: Require a present height, unit cm, a whole number and range 1 through 10 inclusive. For example, 7 cm is accepted; blank is missing, 60 mm has the wrong unit and 11 cm is outside the range. Fractional values such as 4.5 cm also fail. The contract defines accepted inputs; it does not list only the fitting records or establish prediction correctness.

Repeating Efficiently

Without opening the old lesson, explain one key idea from Repeating Efficiently. Give an example and a mistake that the idea helps you avoid.

Look for: Trace a loop by testing its condition before every repetition

Reference example, not the only acceptable wording: Tests: 2 true, 4 true, 6 true, 8 false. Three additions are three iterations. The loop reports 8; the fourth test does not run the body.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

On Monday a team predicts 'rain on Tuesday'. On Tuesday they count a damp playground as success, although sprinklers ran. Write an observation rule that could have been agreed before Tuesday.

Concept to explain

A prediction is a fallible output supported by clues.

Term · Meaning

Term: prediction

Meaning: estimate about an unknown outcome

Term: evidence

Meaning: information that supports or challenges a claim

Term: fact

Meaning: claim established by direct or reliable checking

Term: ground truth

Meaning: reference outcome used to evaluate a prediction

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach comparison as an evidence procedure. The ramp and drying records are fictional; do not present either predictor as a scientific law. Distinguish predictor inputs from outcome measurements, correct rule execution from outcome agreement, and an unassessed case from a failure. Keep observation quality open to documented review.

Readiness answer

It remains unassessed. Obtain a reliable distance record for the same trial, with units and the rule defining far. Repeating the prediction does not provide the missing observation.

Ramp trial cards A through F with heights and separate distance cards from the lesson

A record sheet with trial ID, model version, input, prediction, observation, outcome category and status

Drying records P through S from W4; paper only and no personal data

End goal for the pictorial case

By the end, I can keep predicted and observed outcomes separate and audit a disagreement.

This added worked case illustrates the week's main idea. Retain all original and connected topics;

schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

INPUT available evidence

PROCESS use a rule or model to estimate an unseen outcome

OUTPUT prediction that must remain separate from verified fact

CORE IDEA A prediction is a fallible output supported by clues.

UNDER THE HOOD Prediction estimates a target that is unknown at decision time. Evidence supports the estimate; ground truth is the later or independently established outcome. Treating predictions as labels can create self-confirming records.

Draw a fresh example with all three stages:

Every step connects a claim to observable evidence.

CASE A rain prediction uses weather clues, but the later observation of rain or no rain is the fact that evaluates it.

STEP 1 — OBSERVE available evidence

STEP 2 — TRACE use a rule or model to estimate an unseen outcome

STEP 3 — CONCLUDE prediction that must remain separate from verified fact

MODEL REASONING Mark model or rule outputs as predictions, observable clues as evidence, and independently checked outcomes as facts. A prediction becomes evaluated against a fact; it does not become a fact merely because time passes.

DO NOT OVERCLAIM Do not copy a model prediction into the answer column and later call it ground truth.

What new evidence could change this answer?

Concrete teaching case

A weather helper predicts tomorrow's rain.

Is this a fact about tomorrow? How can it be checked?

Exact explanation to model

It is a prediction. Record it now and compare with a predefined rain observation tomorrow. Useful predictions are not guarantees.

Explain the added worked example

A prediction states what a model outputs. An observation records what was measured or happened. To check a prediction, keep both records and compare the same case under the same outcome rule. Writing a prediction into the observation column would make the check circular.

Use a fictional ramp investigation. Before each trial, Ramp1 predicts far for a ramp height of at least 6 cm and near otherwise. Its allowed heights are whole centimetres from 1 through 10. After a toy rolls, a learner measures its travel distance. The outcome rule is far for a distance of at least 4 floor units and near otherwise. Ramp height and travel distance are different quantities.

The values in this lesson are invented practice records, not a physical law about ramps. Ramp1 is a fixed paper model for learning how to check predictions. Each trial has an ID so that its height, prediction and later measurement stay together. Keep the model version and record the prediction before opening the observation card.

For B, the input 6 cm produces the prediction far. The later distance of 3 units produces the observed category near. This is a mismatch. Retain far as the original prediction and near as the observed category. Changing the old prediction to near would hide the mistake rather than improve the original result.

For A and C, the categories match. These two matches provide evidence about these trials. They do not show that every future prediction will be correct. Even a correctly computed prediction can disagree with an observation because following the model rule and matching the world are different checks.

Key terms

Term · Meaning

Term: Prediction

Meaning: A model output recorded before consulting the comparison outcome

Term: Observation

Meaning: A recorded measurement or event used as evidence

Term: Outcome rule

Meaning: The agreed method for turning observed evidence into the category being checked

Term: Mismatch

Meaning: A prediction that differs from an established comparison outcome

Term: Unassessed

Meaning: A case whose prediction has not yet been judged against adequate outcome evidence

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

D: height 5 cm; predicts near · Distance missing · Unassessed; request the trial record

D: height 5 cm; predicts near: E: height 7 cm; predicts far

Distance missing: Two records say 3 and 4 units

Unassessed; request the trial record: Unassessed; investigate the conflict

D: height 5 cm; predicts near: F: height 6 cm; predicts far

Distance missing: Verified distance exactly 4 units

Unassessed; request the trial record: Match; at least 4 includes 4

For E, 3 units means near and 4 units means far. The disagreement can change the judgment. Check the trial ID, original measurement and recording procedure. Do not choose the distance that makes Ramp1 look better. If the conflict cannot be resolved, retain both reports and mark the comparison unassessed.

An observation is evidence, not an automatic guarantee of truth. A wrong trial ID, unclear unit or copying error can make the comparison unreliable. If a supported correction is made, retain the original entry, the correction, its source and the reason. Remeasuring a new trial does not reveal what happened in the old one.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Prediction Is Not Fact

A fictional drying trial uses a fixed rule and later measures what actually happened.

The facts we will use

Model D predicts slow when moisture score is at least 5; quick otherwise.

Observation policy: slow if drying takes above 8 minutes; quick otherwise. Trial T has score 4 and measured time 9 minutes.

Prediction Is Not Fact

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Prediction input

T has moisture score 4.



2. Predict

4 is below 5, so D predicts quick.



3. Observe later

Measured time is 9 minutes; above 8 means observed slow.



4. Compare evidence

Prediction quick disagrees with observation slow.

CHANGE ONE THING

Keep score 4 but measure exactly 8 minutes.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Prediction input

What happens and why: T has moisture score 4.

Step: 2. Predict

What happens and why: 4 is below 5, so D predicts quick.

Step: 3. Observe later

What happens and why: Measured time is 9 minutes; above 8 means observed slow.

Step: 4. Compare evidence

What happens and why: Prediction quick disagrees with observation slow.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The moisture score generates a prediction; it is not the observed drying time. The later measurement supplies separate evidence under the stated observation policy. Save both values and their units. Altering the recorded time to match D would destroy the evaluation rather than repair the model.

What this example does not establish

One failure does not prove every prediction is wrong; one agreement does not establish reliability on future trials.

Change one thing in the pictured case

Change: Keep score 4 but measure exactly 8 minutes.

Prediction: The observed label becomes quick and agrees with the prediction.

Reason: Above 8 is strict: equality is quick under this policy.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: A weather helper predicts tomorrow's rain.

Write the question to explain: Is this a fact about tomorrow? How can it be checked?

Reveal after predictions: It is a prediction. Record it now and compare with a predefined rain observation tomorrow. Useful predictions are not guarantees.

Keep for an independent attempt: A model predicts rain tomorrow. Tomorrow is dry. Was the prediction an observation of tomorrow? Explain what new evidence settles the result.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

A predicts near, observed near, match; B far, observed near, mismatch; C far, observed far, match. D predicts near with missing distance; E predicts far with conflicting 3 and 4; both unassessed. Two matches out of three assessed, two unassessed. Exit: changing B after reading the outcome hides a mismatch; retain its original far prediction and observed near outcome, and identify any revised model separately.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start 0 to 7 minutes

Teacher action: Retrieve why a missing observation prevents judgment even when a prediction is available.

Stage: 7 to 20 minutes

Teacher action: Model A, B and C with separate height and distance cards. Record predictions before revealing distances. Explain that the lesson shows a worked retrospective example.

Stage: 20 to 33 minutes

Teacher action: Complete W1 and W2. Require both numerical rules and include the exact-distance boundary at 4.

Stage: 33 to 45 minutes

Teacher action: Discuss D and E and complete W3. Ask which evidence could establish the old trial

outcome and why a new trial cannot do so.

Stage: Session 2 start 0 to 8 minutes

Teacher action: Retrieve input, prediction, observation and outcome category with four labelled spaces.

Stage: 8 to 23 minutes

Teacher action: Complete the guided A-through-E results record. Make assessed and unassessed counts visible before forming a result fraction.

Stage: 23 to 38 minutes

Teacher action: Collect W4 independently. Watch for the different greater-than boundary at 10 minutes and the missing S outcome.

Stage: 38 to 45 minutes

Teacher action: Discuss W5 and the exit. Preserve old predictions, distinguish documented observation corrections, and connect model revision to new reserved tests.

Reduce help deliberately

Completion task: Trial F has height 6 cm and verified travel distance exactly 4 units. Apply Ramp1 and the outcome rule. A partner says exactly 4 is near because far means greater than 4. Use the stated rule to resolve the disagreement.

Teacher check: Ramp1 predicts far at height 6. The observation is far because the stated outcome rule is at least 4, including equality. The prediction matches. Greater than 4 would be a different rule and must not replace the agreed one after viewing the result.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Predictions and observations. Proposed explanation: “Running the rule correctly proves the prediction correct”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Rule execution and agreement with the observed outcome are separate checks.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Which field changes the observation label, and which field changes the model prediction?

Reasoned teacher answer

Time changes the observation; moisture score changes the prediction. At score 4 and time 8, both labels are quick.

Changed-case reasoning: The observed label becomes quick and agrees with the prediction. Above 8 is strict: equality is quick under this policy.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Predict first, test honestly, and explain what the evidence means.

State your prediction before the result is known.

Complete the task: Separate prediction, evidence, and fact in four scenarios.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

CLAIM TO REPAIR A prediction is a fact because a computer produced it.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Running the rule correctly proves the prediction correct: Rule execution and agreement with the observed outcome are separate checks.

A missing outcome is a success or a failure: Mark it unassessed and report the missing count alongside assessed results.

Every recorded observation must be true: Check the trial identity, units and evidence; document supported corrections.

A new trial repairs the old observation: A new trial has its own ID and result; the old outcome may remain unknown.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

It is a prediction. Record it now and compare with a predefined rain observation tomorrow. Useful predictions are not guarantees.

Independent question

A model predicts rain tomorrow. Tomorrow is dry. Was the prediction an observation of tomorrow? Explain what new evidence settles the result.

Expected independent answer

No. A prediction estimates an unknown outcome. Tomorrow's actual weather settles whether it was

correct. A plausible prediction can still be wrong.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “A prediction is a fact because a computer produced it.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

For example, record precipitation observed at the school weather station between 00:00 and 24:00 Tuesday above its stated detection threshold. Damp ground alone has another possible cause. Save prediction time, location and observation method before evaluating the result.

Explanations for the additional workbook examples

W1 Separate inputs from observed outcomes

For Ramp1 trial B, height is 6 cm and the later travel distance is 3 units. State the input, prediction, observation, observed category and comparison status. Explain why the prediction rule and outcome rule use different measurements.

Input height 6 cm; prediction far because 6 is at least 6; observed distance 3 units; observed category near because 3 is below 4; mismatch. Height is available to the model before the trial, while travel distance is measured afterward to check its prediction.

W2 Check a boundary without changing the rule

Trial F has height 6 cm and verified travel distance exactly 4 units. Apply Ramp1 and the outcome rule. A partner says exactly 4 is near because far means greater than 4. Use the stated rule to resolve the disagreement.

Ramp1 predicts far at height 6. The observation is far because the stated outcome rule is at least 4, including equality. The prediction matches. Greater than 4 would be a different rule and must not replace the agreed one after viewing the result.

W3 Keep incomplete evidence visible

D predicts near but has no distance record. E predicts far but its two distance records say 3 and 4 units. State each comparison status, request useful evidence, and explain why neither guessing nor a new trial can establish the old trial's outcome.

Both are unassessed. For D request the original distance record for D with units. For E inspect the trial IDs, original measurement and recording procedure to resolve 3 versus 4. Guesses supply no observation; a new trial produces a new outcome. Retain unresolved evidence if the old result cannot be established.

W4 Transfer the check to drying predictions

In a fictional paper test, Dry1 predicts slow for a moisture score of at least 7, quick otherwise. Allowed scores are whole numbers 1 through 10. The observed outcome is slow if drying time is greater than 10 minutes, quick otherwise. Records are P: score 6, time 12; Q: score 7, time 10; R: score 9, time 14; S: score 5, time missing. Record all predictions, observed categories and comparison statuses. Summarise matches out of assessed cases, name the unassessed count, and limit the conclusion.

P predicts quick, observed slow, mismatch. Q predicts slow, observed quick because exactly 10 is not greater than 10, mismatch. R predicts slow, observed slow, match. S predicts quick, observed outcome unavailable, unassessed. One match out of three assessed cases, with one unassessed. This describes

the assessed cases and does not establish the result for S or future trials.

W5 Correct a report without erasing evidence

After seeing B's observed near category, a learner replaces Ramp1's far prediction with near. They call the new entry a corrected observation and report a perfect result for A through C. Explain the two record-keeping errors and write a defensible replacement report.

They changed a prediction after seeing the outcome and mislabelled that change as an observation correction. Restore the original far prediction for B and retain the observed near category. Report two matches out of three assessed Ramp1 trials A through C, including B's mismatch. Any revised model needs a new version and new reserved cases for evaluation.

Assess independent transfer

Selected case: In a fictional paper test, Dry1 predicts slow for a moisture score of at least 7, quick otherwise. Allowed scores are whole numbers 1 through 10. The observed outcome is slow if drying time is greater than 10 minutes, quick otherwise. Records are P: score 6, time 12; Q: score 7, time 10; R: score 9, time 14; S: score 5, time missing. Record all predictions, observed categories and comparison statuses. Summarise matches out of assessed cases, name the unassessed count, and limit the conclusion.

Selected case answer

P predicts quick, observed slow, mismatch. Q predicts slow, observed quick because exactly 10 is not greater than 10, mismatch. R predicts slow, observed slow, match. S predicts quick, observed outcome unavailable, unassessed. One match out of three assessed cases, with one unassessed. This describes the assessed cases and does not establish the result for S or future trials.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with

a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Helpful Classroom Sorter

Label the sorter's suggested bin as a prediction until a human confirms the final placement.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two mechanisms with the same visible output and explain what evidence distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 18: Confidence Is Not Correctness. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Use separate coloured cards for predictions and observations and pair them by trial ID. Read at least and greater than aloud on number lines. Extension: investigate how selective missing records could make the assessed cases an incomplete picture of model performance.

W4 is secure with quick/slow/slow/quick predictions, slow/quick/slow/unknown outcomes, two mismatches, one match and one unassessed case. Require one of three assessed and limited claims. After support retry scores 5, 8, 4 with times 9, 10, missing: quick/slow/quick predictions, quick/quick/unknown outcomes; one match, one mismatch, one unassessed.

Week 18 distinguishes confidence from correctness and asks what evidence could support a confidence claim.

Plan the delayed return

Suggested return points in this level: ai-1/18, ai-1/20, ai-1/23. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

In sorter trials, record predicted categories before collecting reference evidence, then retain disagreements for investigation.

7. Prepare, teach and assess

Teaching target: Separate a recorded prediction from the observation used to check it

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Trial F has height 6 cm and verified travel distance exactly 4 units. Apply Ramp1 and the outcome rule. A partner says exactly 4 is near because far means greater than 4. Use the stated rule to resolve the disagreement.

Reasoned practice answer: Ramp1 predicts far at height 6. The observation is far because the stated outcome rule is at least 4, including equality. The prediction matches. Greater than 4 would be a different rule and must not replace the agreed one after viewing the result.

Check every learner before discussion: Use the worked case for Predictions and observations. Proposed

explanation: “Running the rule correctly proves the prediction correct”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Rule execution and agreement with the observed outcome are separate checks.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Keep score 4 but measure exactly 8 minutes. Require a prediction and a reason before comparing. Expected result: The observed label becomes quick and agrees with the prediction. Reason: Above 8 is strict: equality is quick under this policy.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner’s explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.